

Domain Walls and Ising-Bloch Transitions in Parametrically Driven Systems

Germán J. de Valcárcel,* Isabel Pérez-Arjona, and Eugenio Roldán

Departament d'Òptica, Universitat de València, Dr. Moliner 50, 46100-Burjassot, Spain

(Received 27 October 2000; revised manuscript received 27 June 2002; published 30 September 2002)

Parametrically driven systems sustaining sech solitons are shown to support a new kind of localized state. These structures are walls connecting two regions oscillating in antiphase that form in the parameter domain where the sech soliton is unstable. Depending on the parameter set the oppositely phased domains can be either spatially uniform or patterned. Both chiral (Bloch) and nonchiral (Ising) walls are found, which bifurcate one into the other via an Ising-Bloch transition. While Ising walls are at rest Bloch walls move and may display secondary bifurcations leading to chaotic wall motion.

DOI: 10.1103/PhysRevLett.89.164101

PACS numbers: 05.45.Xt, 42.65.Tg, 47.35.+i, 75.60.Ch

The periodic forcing of dissipative, dispersive nonlinear systems may lead to the excitation of sustained waves oscillating at half the driving frequency, a phenomenon known as parametric resonance. One of its paradigms is the excitation of surface waves in liquids vertically vibrated on horizontal plates. Apart from supplying energy to the system, the periodic forcing breaks the continuous time translation invariance of the undriven system down to the discrete one $t \rightarrow t + 2\pi/\omega_f$, where ω_f denotes the angular frequency of driving. Alternatively, in the absence of driving, the (damped) oscillations can have an arbitrary phase, whereas forced waves *see* an external clock and their phase tends to lock to it: the phase invariance of the undriven oscillations is broken. In particular, when the frequency of oscillations is half that of driving, as in parametric resonance, symmetry considerations [1] allow one to conclude that there exist two dynamically equivalent preferred phases differing by π . In spatially extended systems this equivalence allows the antiphase oscillation of two neighboring regions (domains), which appear separated by a so-called wall. Two types of walls, namely, Ising and Bloch walls, exist which differ in the way the phase of oscillations changes across them: In an Ising wall the phase displays a discontinuous π jump at a nonoscillating node, whereas in a Bloch wall the phase angle rotates continuously through π . Hence Bloch walls are chiral (have the symmetry of a corkscrew) while Ising walls are not, and this has important effects on the walls dynamics [1,2]. Ising and Bloch walls may bifurcate one into the other via a nonequilibrium Ising-Bloch transition [2,3], similar to the (equilibrium) Ising-Bloch transition of ferromagnets.

Ising walls in the form of hyperbolic tangents (also called kinks or dark solitons) have been observed in the parametrically driven surface waves of liquids in *shallow* channels [4] and in the parametric excitation of the *upper* cutoff mode of a chain of coupled pendula [5]. A key feature of both systems is that the sign of dispersion and nonlinear frequency shift are opposite. On the contrary, when they have the same sign, the localized structures supported by parametrically driven systems are of a

different nature. Now solitons have the form of hyperbolic secants (also called solitary waves or bright solitons) and have been observed in the parametrically driven surface waves of liquids in *deep* channels [6] and in the parametric excitation of the *lower* cutoff mode of a chain of coupled pendula [7]. Both tanh and sech solitons are well understood in the frame of nonlinear Schrödinger models generalized by terms accounting for dissipation and parametric drive [4,5,7,8]. In this Letter we show that parametrically driven systems supporting sech solitons may display domain walls as well. These new structures are predicted to exist within the parameter region where the soliton is unstable against spatially extended modes.

We consider a basic equation widely used for modeling the parametric forcing of cutoff modes in one dimensional systems,

$$\partial_t u = \gamma u^* - (1 + i\theta)u + (i + \delta)\partial_x^2 u + (i - \sigma)|u|^2 u, \quad (1)$$

where u is the slowly varying complex envelope of oscillations and u^* is its complex conjugate. A typical scalar variable h , e.g., the fluid surface displacement from planarity, can be expressed in terms of u as $h(x, t) = \text{Re}[u(x, t) \exp(-i\omega_f t/2)]$. Equation (1) is adimensional and contains four real parameters: (i) the driving strength γ , proportional to the amplitude of forcing; (ii) the detuning θ , proportional to the difference between $\omega_f/2$ and the frequency of the cutoff mode; (iii) the wave-number-dependent damping coefficient δ ; and (iv) the nonlinear damping coefficient σ . Parameters δ and σ are non-negative, and γ is taken as non-negative without loss of generality. Note that the term γu^* breaks the continuous phase symmetry $u \rightarrow e^{i\phi} u$ of the undriven ($\gamma = 0$) system down to the discrete one $u \rightarrow -u$, so that each state has a single equivalent one, of opposite phase, as commented.

For $\delta = \sigma = 0$, Eq. (1) is a parametrically driven, damped nonlinear Schrödinger (PDNLS) equation. It is used to model parametrically excited surface waves in long channels [8], parametrically driven chains of

coupled pendula [7,9], parametric waves in plasma [10], phase-sensitive amplification in optical systems [11], and magnetization waves in parametrically pumped easy-plane ferromagnets [12]. It also governs the dynamics [12] of parametrically driven, damped sine-Gordon systems [13].

The terms proportional to δ and σ in Eq. (1) are considered here as a perturbation of the PDNLS equation. They represent actual dissipation mechanisms [14,15], although they are usually neglected in the studies of parametric resonance. We keep these terms since they have important effects on the solutions here analyzed [16].

Before reporting on the new states, let us briefly comment on the known solutions to the PDNLS equation ($\delta = \sigma = 0$). (i) For $\theta < 0$ the trivial state $u = 0$ becomes unstable at $\gamma = 1$ giving rise supercritically to roll solutions [17]. (ii) For $\theta > 0$ the trivial state undergoes a subcritical bifurcation at $\gamma = \gamma_\theta \equiv \sqrt{1 + \theta^2}$, leading to a sech soliton for $1 < \gamma < \gamma_\theta$ [8,9,12]. For $\gamma > \gamma_\theta$ the soliton is destabilized since the trivial state (to which the solitons asymptotically tends) is unstable and spatially extended modes are amplified. In this region ($\gamma > \gamma_\theta$ and $\theta > 0$), stable roll solutions $u_{\text{roll}}(x) \approx \pm e^{i\varphi_0}(\rho_0 + \rho_1 \cos k_c x)$, with $\cos 2\varphi_0 = \gamma^{-1}$, $\rho_0^2 = 3\theta/5 + \sqrt{\gamma^2 - 1}$, $\rho_1^2 = 4\theta/15$, and $k_c^2 = \theta + 2\sqrt{\gamma^2 - 1}$, exist [18]. Note that $u_{\text{roll}}(x)$ is not symmetric with respect to $u = 0$ because of the bias term ρ_0 . We argue that, given the broken phase invariance of Eq. (1), stable walls joining the two ($\pm$) roll states should exist for $\gamma > \gamma_\theta$ and $\theta > 0$. Clearly, different to the usual ($\tanh$) domain walls, these new solutions should connect two patterned states.

In order to determine the nature of the sought walls we introduce a chirality parameter $\chi = \text{Im}[u^*(x_0)\partial_x u(x_0)]$, where x_0 is the position of the wall core (where $|u|$ is closest to zero). For Ising (Bloch) walls $\chi = 0$ ($\chi \neq 0$). Since Eq. (1) is invariant under spatial translations ($x \rightarrow x + x'$) and spatial reflections ($x \rightarrow -x$), if a Bloch wall $u_1 = u_B(x - x_0, t)$ exists with chirality χ_1 another Bloch wall $u_2 = u_B(x_0 - x, t)$ also exists and has chirality $\chi_2 = -\chi_1$. This is important since a generic property of domain walls (at least of the $\tanh$ type) in nongradient systems—such as Eq. (1)—is that they move with a velocity proportional to their chirality [2], so that Bloch walls with opposite chiralities should move in opposite directions, and Ising walls ($\chi = 0$) should be at rest.

We show next the results of our numerical study of Eq. (1) [19]. Since periodic boundary conditions were used, not one but two domain walls had to be excited. Nevertheless, the chosen integrating window length L allowed both walls to be separated enough so that there was no interaction between the two walls [20].

Bloch walls in the PDNLS equation.—For $\delta = \sigma = 0$ only Bloch walls are found, of which Fig. 1 (upper row) is an example. The domain wall nature of this structure is evident in Fig. 1(a) where $\text{Re}(u)$ is shown to consist of two

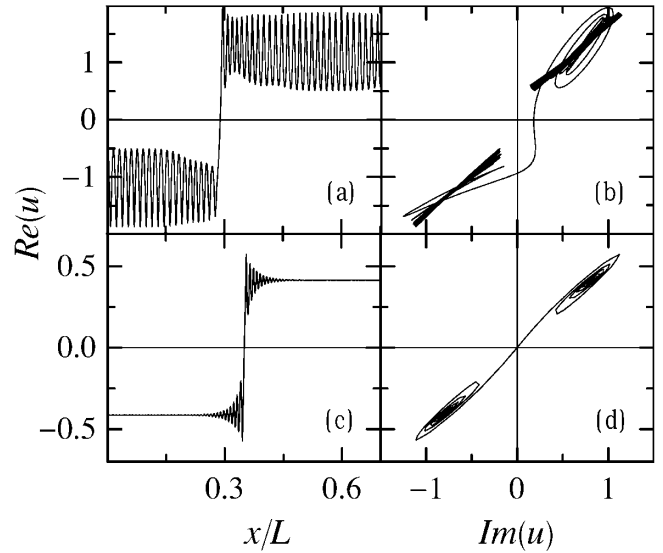


FIG. 1. Upper (lower) row corresponds to a Bloch (Ising) wall. Parameters are $\gamma = 2.0$, $\theta = 0.8$, $\sigma = \delta = 0$ for the Bloch wall, and $\gamma = 1.25$, $\theta = 0.1$, $\delta = 0$, $\sigma = 0.01$ for the Ising wall.

rolls with opposite signs (phases) joined by a wall. The fact that these are Bloch walls is evident in Fig. 1(b) where u is seen to be different from zero everywhere, so that its phase varies smoothly from one roll to the other. As expected these Bloch walls move although their motion is not "clean": walls are usually wandering around a fixed location in an irregular way. These walls exist within region B of Fig. 2(a). Region B is bounded from below by the line $\gamma = \gamma_\theta$ (below this line walls destabilize leading to chains of bright solitons in a way similar to [18]) and from above. As the upper boundary is approached from region B the spatial modulations next to the wall, see Fig. 1(a), increase and approach the complex zero. At the boundary these deep modulations become so close to the complex zero that new defects are nucleated. The above mechanism affects in turn each of the new defects and propagates across the entire pattern, eventually leading to a weakly turbulent, unbiased roll pattern.

Ising and Bloch walls in the perturbed PDNLS equation.—The effects of a nonzero δ or a non-null σ in Eq. (1) are similar and fourfold: (i) the region of existence of domain walls is enlarged; (ii) the spatially uniform solutions (always unstable in the PDNLS equation [18]) stabilize within some parameter region and usual domain walls connecting spatially homogeneous states are excited, besides the spatially modulated domain walls; (iii) besides the Bloch walls reported above, Ising walls also exist and bifurcate into Bloch walls via an Ising-Bloch transition; and (iv) Bloch walls display secondary instabilities affecting their motion, eventually leading to chaotic behavior.

Figure 2 summarizes points (i)–(iii) above by showing the domains of existence of Ising and Bloch walls on the

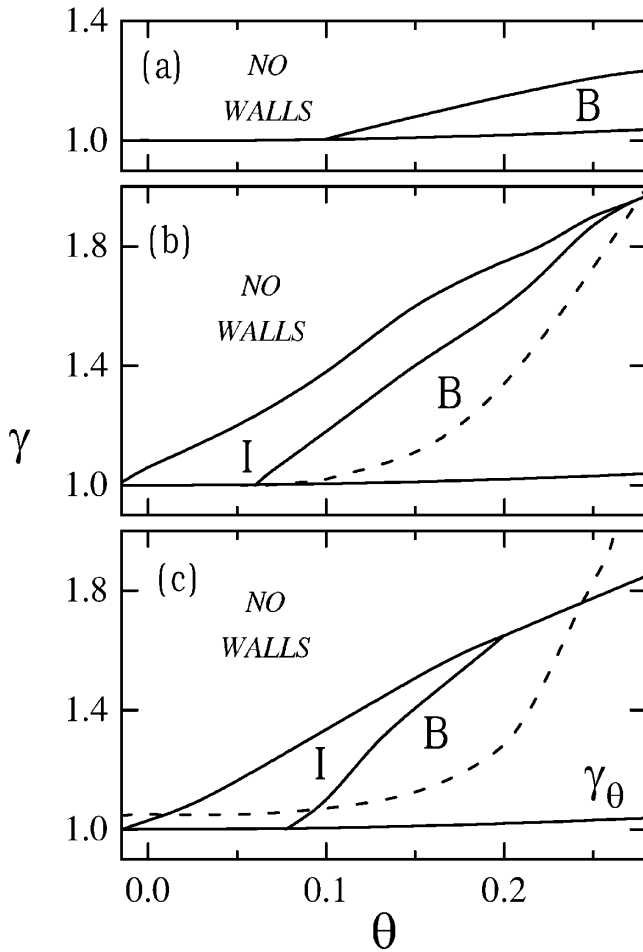


FIG. 2. Numerically determined existence regions of Ising (I) and Bloch (B) walls. (a) corresponds to the PDNLS equation ($\sigma = \delta = 0$). (b) $\delta = 0.01$, $\sigma = 0$. (c) $\delta = 0$, $\sigma = 0.01$. The nontrivial homogeneous state is stable above the dashed line and the trivial state $u = 0$ is unstable above line γ_θ , only indicated in (c).

plane $\gamma - \theta$ for $\{\delta = 0.01, \sigma = 0\}$, Fig. 2(b), and $\{\delta = 0, \sigma = 0.01\}$, Fig. 2(c). The line that separates regions *I* from *B* corresponds to the Ising-Bloch transition. Outside regions *I* and *B* walls destabilize through the same mechanisms described above for the PDNLS equation. Also depicted (dashed line) is the stability boundary of the spatially homogeneous states as predicted by their linear stability analysis: to the left of the dashed line the homogeneous steady states are stable and can form domain walls, which can be of either Ising or Bloch type. To the right of the dashed line the homogeneous states destabilize towards roll solutions similar to u_{roll} and domain walls (of Ising or Bloch character) similar to those found in the PDNLS equation form. An example of an Ising wall connecting two spatially homogeneous states is shown in the bottom row in Fig. 1. We have checked that by increasing the values of σ and δ (keeping them small) the region where domain walls exist is enlarged, as well as the region where Ising walls are found.

In Fig. 3 we show a typical bifurcation diagram obtained for $\{\gamma = 1.3, \delta = 0, \sigma = 0.01\}$. By increasing θ from the Ising region the Ising-Bloch transition occurs at $\theta = 0.134$ where the Ising wall (at rest) bifurcates into two Bloch walls of opposite chirality that move with opposite velocities. Which one of the two Bloch walls is excited is selected by numerical noise. One can also force the direction of motion by giving a small chirality χ_0 to the wall as an initial condition, the sign of χ_0 controlling the direction of the drift. At $\theta = 0.185$ a secondary bifurcation takes place, Fig. 3, where a new, oscillatory motion is superimposed to the already constant drift of the walls; see Fig. 4(a). Figure 4(b) displays a projection onto the plane $\chi - v$ of the attractor associated to Fig. 4(a) showing a clear correlation between these two quantities even under oscillatory motion. Also shown in Fig. 4(b) is the symmetric attractor corresponding to the oppositely moving wall, with an accordingly opposite chirality. Finally, by further increasing θ oscillations become more and more pronounced, Fig. 3, until the velocity minima become virtually zero. In this situation the two symmetric attractors [analogous to those in Fig. 4(b)] glue and merge, the wall dynamics alternates between these two lobes, Fig. 4(c), and a single chaotic attractor is formed, Fig. 4(d).

We finally note that other scenarios, related to the one reported here, have also been found and will be reported elsewhere since they are not essential to our findings. We just remark that when diffusion is considered ($\delta \neq 0$) the bifurcation diagram is similar to the one found in the anisotropic XY model of statistical physics weakly perturbed by nongradient terms [1].

In conclusion we have shown that parametrically driven systems exhibiting sech solitons can also support tanh-like states in the high parametric gain region where the sech solitons are unstable. This new solution is a

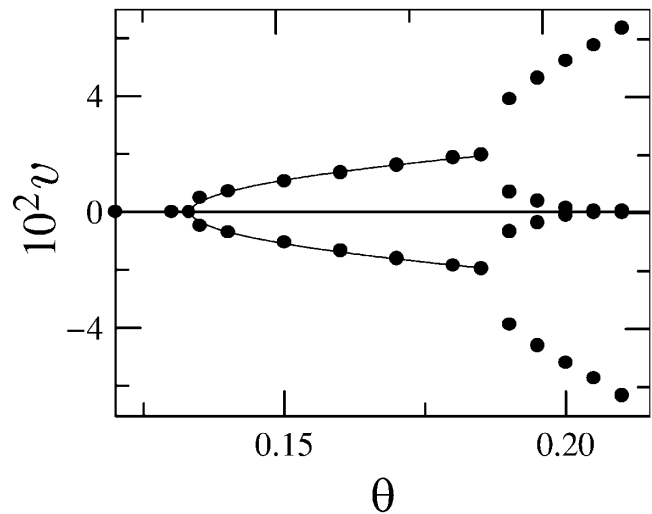


FIG. 3. Wall velocity (v) as a function of detuning (θ) for $\gamma = 1.3$, $\delta = 0$, $\sigma = 0.01$. The curve is a parabolic fit.

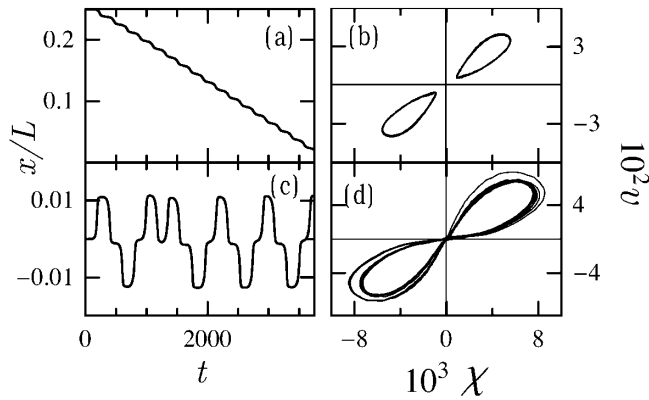


FIG. 4. Evolution of the wall location (left column) and attractor's projection onto the plane velocity vs chirality (right column). Parameters as in Fig. 3 with $\theta = 0.2$ (a),(b) and $\theta = 0.25$ (c),(d).

domain wall connecting two oppositely phased states, both spatially homogeneous or patterned, which exhibits an Ising-Bloch transition. Bloch walls move and exhibit secondary bifurcations eventually leading to chaotic motion. A clear relation between the chirality associated to the wall core and the velocity of the whole pattern has been reported which means that the global dynamics of the pattern is controlled by the local properties of the core of the defect. Long channels of deep liquid and chains of coupled pendula parametrically forced at twice the frequency of the fundamental cross mode and the lower cutoff mode, respectively, could be ideal systems to test the phenomena reported here.

We thank V.J. Sánchez-Morcillo, M. San Miguel, and F. Silva for fruitful discussions and J.C. Jackson for his reading of the manuscript. Financial support from the Spanish Government (DGES Project No. PB98-0935-C03-02) and from ESF network PHASE is acknowledged.

*Electronic address: german.valcarcel@uv.es

- [1] P. Coullet and K. Emilsson, *Physica (Amsterdam)* **61D**, 119 (1992).
- [2] P. Coullet *et al.*, *Phys. Rev. Lett.* **65**, 1352 (1990).
- [3] D. Michaelis *et al.*, *Phys. Rev. E* **63**, 066602 (2001).

- [4] B. Denardo *et al.*, *Phys. Rev. Lett.* **64**, 1518 (1990).
- [5] B. Denardo *et al.*, *Phys. Rev. Lett.* **68**, 1730 (1992).
- [6] J. Wu, R. Keolian, and I. Rudnick, *Phys. Rev. Lett.* **52**, 1421 (1984).
- [7] B. Denardo, Ph.D. thesis, University of California, Los Angeles, 1990 (unpublished).
- [8] J.W. Miles, *J. Fluid Mech.* **148**, 451 (1984); C. Elphick and E. Meron, *Phys. Rev. A* **40**, 3226 (1989).
- [9] N.V. Alexeeva, I.V. Barashenkov, and G. P. Tsironis, *Phys. Rev. Lett.* **84**, 3053 (2000).
- [10] V.E. Zakharov, S.L. Musher, and A.M. Rubenchik, *Phys. Rep.* **129**, 285 (1985).
- [11] A. Mecozzi *et al.*, *Opt. Lett.* **19**, 2050 (1994); S. Longhi, *Phys. Scr.* **56**, 611 (1997); V.J. Sánchez-Morcillo *et al.*, *Opt. Lett.* **25**, 957 (2000).
- [12] I.V. Barashenkov, M.M. Bogdan, and V.I. Korobov, *Europhys. Lett.* **15**, 113 (1991).
- [13] N. Grønbech-Jensen, Y.S. Kivshar, and M.R. Samuelsen, *Phys. Rev. B* **47**, 5013 (1993); R. Grauer and Y.S. Kivshar, *Phys. Rev. E* **48**, 4791 (1993).
- [14] S.T. Milner, *J. Fluid Mech.* **225**, 81 (1991).
- [15] G.P. Agrawal, *Nonlinear Fiber Optics* (Academic Press, San Diego, CA, 1995); A.M. Dunlop, W.J. Firth, and E.M. Wright, *Opt. Commun.* **138**, 211 (1997); Y.S. Kivshar and B. Luther-Davies, *Phys. Rep.* **298**, 81 (1998).
- [16] Our case is similar to [17], where $\sigma = 0.05$ had to be used in order to reproduce the experimentally observed secondary instabilities of parametric surface waves.
- [17] W. Zhang and J. Viñals, *Phys. Rev. Lett.* **74**, 690 (1995).
- [18] S. Longhi and A. Geraci, *Appl. Phys. Lett.* **67**, 3060 (1995).
- [19] A split-step algorithm was used. Spatial grids from 1024 to 8192 points were used in order to check convergence. Results reported here were obtained with an integrating window length $L = 300$ (other values were also used) and the temporal step was lowered to $\delta t = 10^{-3}$ in order to yield δt -independent results.
- [20] In order to check this we also made integrations for a localized parametric driving $\gamma = \gamma_0 \exp[-(x/\Delta x)^8]$ with $\Delta x = 0.45L$, which is flat around $x = 0$ and null close to the border ($x = \pm L/2$) of the integration region. Single domain walls could be excited obtaining identical results as with spatially uniform driving. This localized driving is not just a numerical trick as it describes, e.g., a light beam with finite transverse extension injected into a nonlinear optical system.